with an embedded VCSEL driver. The VL53L8 is capable of delivering 16 or 64 discrete ranging zones with stable and accurate improved ranging. It can also support indoor/outdoor detection and smart-focus bracketing, as well as consumer lidar—where depth mapping is required. The VL53L8 is IEC 60825-1 Class 1 certified and is fully eye-safe during normal applications.

STMicroelectronics
https://www.st.com

Low-current detect switches

C&K has introduced a series of ultra-low-current (ULC) detect switches. These switches have a single-pole single-throw (SPST) design and are

tested for over 50,000 cycles, 75gf actuation force, and right angle termination. The SDS ULC series comes in multiple contact arrangements suitable for different use cases. These have silver and gold plated tips for better resistance to corrosion and are suitable for operation in temperature from –40°C to +85°C. The SPST switches can be employed for a wide variety of applications, such as IoT, medical, networking, and smart metering.

C&K switches
https://www.ckswitches.com

1Gbit Ethernet supporting cables

The new M12 cables from CUI Devices can support up to 1Gbit Ethernet. The new cables are offered in multiple configurations of male or female plug versions and come in various contact position options, including 3, 4, 5, 8, and 12-pin configurations. The cables are rated for 1.5A to 4A current and

can carry a maximum voltage of 30V to 250V DC. These come in thicknesses ranging from 26 to 22AWG. The cables are waterproof and come with an IP67 rating, making them suitable for use in harsh industrial setups

CUI Devices
www.cuidevices.com

Next-generation memory

Samsung's Memory Semantic SSD integrates DRAM memory and storage. It employs Compute Express Link (CXL) interconnect technology and

an integrated DRAM cache to provide up to 20x boost in random read speed and latency when used in applications of AI and ML. Memory Semantic SSDs support petabyte storage to improve storage capacity. This results in low power consumption and aids in high server utilisation. The SSDs are ideal for a wide variety of AI and ML workloads that require very fast processing of larger data sets.

Samsung
https://www.samsung.com

SMD antenna for GNSS

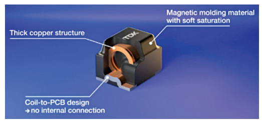
Agosti SR4G080 from Antenova is a surface-mounted device (SMD)

antenna for GNSS applications. It measures just 9.0mm × 5.8mm × 1.7mm and provides omnidirectional performance. The antenna is designed to integrate and synchronise with the other antennas on the same device. The SMD antenna is a corner placement antenna, which offers designers some additional layout options. It is designed to be placed on a corner of the PCB and utilises the PCB surface around the antenna as a ground plane to radiate signal.

Antenova
https://www.antenova.com

Automotive-grade power inductors

CLT32 is a series of high-performance, automotive-grade power inductors from TDK Corp. They provide high-performance electrical values in a small pack-



age. These devices target advanced driver assistance systems (ADAS) and autonomous driving (AD) power management ICs (PMICs) and DC/DC converters with high clock rates and switching frequencies up to 10MHz. They are designed with a solid copper coil moulded over a ferromagnetic plastic compound. The solid copper coil enables these devices to achieve an exceptionally low RDC value, keeping losses to a minimum. The CLT32 are suitable for use in safety-critical automotive applications that use high-performance processors, which require currents in the double-digit ampere range.

TDK Corp
https://www.tdk-electronics.tdk.com

EMBEDDED SYSTEMS

Motherboard with 12th-Gen Intel processor

MBB-1000 ATX motherboard from IBASE Technology is powered by the 12th-Gen Intel Core processors (formerly Alder Lake). It supports four 4K displays and features multiple